Abhiram Mullapudi

LEAD DATA SCIENTIST · MACHINE LEARNING FOR CYBER-PHYSICAL INFRASTRUCTURE

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Data science leader with a PhD and six years of industry experience taking machine learning from research to production for critical water infrastructure. I build and ship forecasting, anomaly detection, and optimization systems that run live at water utilities, and the MLOps platforms that keep them running. Maintainer of the industry-standard open-source stormwater modeling stack; author of 10+ peer-reviewed publications.

EXPERIENCE

Lead Data Scientist *Inframark* 2025–PRESENT

- Lead the development of machine-learning solutions for optimizing collection-system and wastewater-treatment-plant operations, from problem framing with operators to production deployment.
- Building scalable ML tooling that makes operational analytics viable for the small and mid-sized utilities that make up most of the market.

Senior Data Scientist *Xylem* 2023–2025

- Led the development of statistical and ML methodologies for time-series filtering and anomaly detection, powering predictive maintenance and operational decision-making across Xylem's digital water products.
- Designed and built a Flyte-based MLOps platform that standardized model development, deployment, and maintenance across product teams.

Hydraulic Control & Optimization Engineer *Xylem* 2020–2023

- Developed a 1D-CNN forecasting model combining NOAA rainfall forecasts with near-real-time flow measurements to predict 24-hour treatment-plant inflow; deployed as an operator-facing dashboard.
- Designed a real-time anomaly detection system that monitors 600+ concurrent sensor streams using symbolic programming and statistical techniques.
- Engineered a 1D-CNN interpolation framework for spatially distributed river-level data, supporting environmental monitoring and regulatory compliance reporting.
- Built ML-powered visualization products, including a public-facing Combined Sewer Overflow event tracker, and maintained the real-time services, ETL pipelines, and databases behind a production network-optimization product.

SELECTED PUBLICATIONS

Identification of stormwater control strategies and their associated uncertainties using Bayesian Optimization
arXiv preprint 2023

pystorms: a simulation sandbox for the design and evaluation of stormwater control algorithms
Environmental Modelling & Software 2023

Deep Reinforcement Learning for the Real Time Control of Stormwater Systems
Advances in Water Resources 2020

PySWMM: The Python Interface to Stormwater Management Model (SWMM)
Journal of Open Source Software 2020

Full list of 10+ publications on Google Scholar.

EDUCATION

Ph.D., Civil Engineering
University of Michigan · 2017–2020

M.Sc.Eng., Civil Engineering
University of Michigan · 2015–2017

B.Tech., Civil Engineering
Amrita Vishwa Vidyapeetham · 2011–2015

SKILLS

PROGRAMMING
Python, SQL, C/C++, Swift, Bash

MACHINE LEARNING
PyTorch, JAX, scikit-learn; time-series deep learning, reinforcement learning, Bayesian optimization, physics-informed ML

STATISTICS & SIGNAL PROCESSING
Probabilistic modeling, uncertainty quantification, anomaly detection, time-series filtering

MLOPS & CLOUD
Flyte, MLflow, AWS, Google Cloud, Azure

DATA ENGINEERING
Real-time streaming pipelines, ETL, backend services

LEADERSHIP & SERVICE

Vice-chair — Emerging Technologies subcommittee, ASCE EWRI Congress

Organizing committee — NeurIPS Gaussian Processes Workshop, 2024

Peer reviewer — 11 journals, incl. *Water Resources Research*

Grand Prize — LIFT Intelligent Water Systems Challenge, 2019

OPEN SOURCE

pyswmm MAINTAINER
Python interface to EPA-SWMM, the industry standard for stormwater modeling.

OpenFloodHub CREATOR
Per-gauge CNN streamflow forecasting, benchmarked against NOAA's models.

flood sim CREATOR
GPU shallow-water flood simulation in the browser.

realClippy CREATOR
macOS PDF reader with a local-LLM chatbot.